

Contact

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www.linkedin.com/in/yermanmerel
(LinkedIn)

engermanmerel.github.io/
(Portfolio)

Top Skills

Engineering

Construction

Model-based Systems Engineering
(MBSE)

Languages

Ingles (Full Professional)

German (Elementary)

Español (Native or Bilingual)

Certifications

Professional Qualification in
Engineering

Honors-Awards

Finalist in the Scientific Initiation
Conference

Yerman M.

Mechanical Engineer | Tunnelling & Mechanical Systems | Field
Engineering, Asset Optimization & Project Execution

Panamá, Panama

Summary

Mechanical & Systems Engineer with 2+ years of international experience in underground infrastructure (TBM tunneling, metro systems) and industrial piping design. Contributed to projects impacting 160,000+ daily users across Peru, Germany, and Panama. Currently leading systems engineering at an early-stage startup, applying MBSE to align technical R&D with scalable business workflows. Core strengths: structural validation, hydraulic analysis, ASME/AISC compliance, and SOP development for high-criticality mechanical systems.

Experience

Herrenknecht AG

Project Engineer

January 2025 - August 2025 (8 months)

Schwanau

Designed and documented the global SOP for race replacement in the Main Drive of EPB TBM machines.

Performed structural calculation validation and functional analysis of mechanical systems, ensuring strength and safety in high-demand industrial environments according to site conditions.

Guaranteed 100% regulatory compliance and documentary traceability, mitigating contractual fine risks for the Lima Metro and Paradise Tunnel projects.

Projects: Hudson River Project (USA), Railway Tunnel Perú

CHINA RAILWAY TUNNEL GROUP CO., LTD.

Industrial Piping Design Engineer Intern

August 2024 - January 2025 (6 months)

Panama City, Panamá

- Designed and validated mechanical coupling systems for dynamic operational environments, ensuring structural integrity and functional reliability under variable load conditions.
- Conducted technical surveys of existing structures to assess load-bearing capacity and implemented targeted structural improvements to meet operational requirements.
- Performed hydraulic analysis of critical utility systems, including compressed air, chilled water, and process water networks — supporting uninterrupted system performance during pre-commissioning.
- Optimized pipe support systems in compliance with ASME and AISC standards, introducing advanced coupling solutions between pipework and structural elements that improved load distribution and reduced maintenance access constraints.
- Developed and applied a pipeline extension methodology based on OSHA, ASME, and Panamanian regulatory standards, ensuring full safety compliance and operational continuity across newly extended network segments.

Education

Universidad Tecnológica de Panamá

Bachelor of Science (B.Sc.), Mechanical Engineering · (January 2018 - August 2024)